

Summary

- Experienced Data Engineer with over 5 years of hands-on expertise in developing and optimizing end-to-end data pipelines across diverse platforms and cloud environments (GCP, AWS, Azure). Proficient in Python, SQL, NoSQL, Apache Spark, specializing in ETL design, implementation, and maintenance using tools like Informatica, Apache Airflow, and custom scripts. Skilled in data modeling, performance tuning, and ensuring data integrity across Oracle, Snowflake, and Redshift. Successfully led migration projects, implemented Delta Lake strategies, and applied machine learning algorithms to healthcare datasets, improving predictive modeling accuracy and supporting clinical decision-making. Strong collaborator, delivering scalable analytics solutions and enhancing operational efficiency through seamless knowledge transfer.

Skills

Programming Languages: Python, Java, Scala

Big Data and Data Processing: SQL, Apache Spark (PySpark, SparkSQL), Hadoop, Apache Beam, Hive, NoSQL (MongoDB), PL/SQL, Machine Learning

Databases: Snowflake, Amazon Redshift, Google BigQuery, Azure SQL, Oracle, Teradata, SQL Server, PostgreSQL, DB2, Netezza

Cloud Services: Azure (Azure Data Factory, Azure Logic Apps, Azure Databricks, Azure Synapse, Blob Storage, Azure SQL), AWS (AWS Glue, S3, AWS Lambda, AWS CloudWatch, AWS IAM, AWS Athena, AWS EC2, AWS EMR), GCP (GCP Dataflow, GCP Dataproc, GCP BigQuery, GCP Dataform, GCP Data fusion, GCP Cloud SQL, GCP Cloud Storage, GCP cloud spanner, GCP Bigtable)

ETL & Orchestration: ETL Pipelines, Apache Airflow, Informatica, dbt, Custom ETL Design & Implementation

DevOps and Automation: CI/CD Pipeline, Docker, Git, Azure DevOps, Jenkins

Other: Power BI, Tableau, Data Warehousing, Data Modeling, API, Data Scraping, DAG, Knime, Microsoft Excel, Agile, JIRA, FHIR/HL7 integration, IRIS for Health

Professional Experience

RESEARCH DATA ENGINEER | AIM AHEAD | Denton, TX

February 2023 - Present

- Spearheaded integration of diverse nationwide datasets including cancer registry (SEER), medical claims (Medicare), state-level health data (HIMSS), and CMS datasets into a centralized GCP data lake using BigQuery and Cloud Storage, ensuring efficient querying and real-time analytics for large-scale health research.
- Developed and deployed advanced analytics workflows on GCP Dataproc using Python, SQL, and MLlib to process and analyze large-scale healthcare datasets, optimizing performance for actionable insights in cancer research.
- Designed and implemented ETL pipelines using Apache Airflow for orchestration, GCP Dataflow, and Python scripts, reducing processing times and enhancing analytical depth for health research initiatives, enabling faster decision-making.
- Built scalable data warehouse solutions on GCP BigQuery, optimizing query performance and ensuring efficient storage of structured and semi-structured healthcare data, supporting real-time analysis of cancer patient data.
- Implemented CI/CD pipelines using GCP Cloud Build, automating deployment and testing to deliver faster, more reliable analytics and improve time-to-insight in cancer research.
- Managed automated pipeline solutions using GCP Cloud Monitoring, ensuring real-time alerts, efficient debugging, and continuous optimization of workflows, ensuring smooth data operations in a health research environment.
- Conducted statistical analyses on Medicare claims to evaluate the effectiveness of medical interventions, applying explainable AI/ML approaches to enhance model transparency for better decision-making by clinicians and researchers.
- Built predictive models using Gradient Boosting Classifier (GBC) and Ordinary Least Squares (OLS) regression, forecasting cancer outcomes, mitigating biases, and improving model fairness in health-related predictions.
- Ensured data accuracy, reliability, and compliance by implementing rigorous quality assurance, validation processes, and data governance standards across all analysis pipelines, aligning with regulatory requirements.
- Integrated FHIR and IRIS for Health standards for seamless interoperability, ensuring accurate and accessible data exchange between healthcare data systems.
- Developed and implemented advanced data visualization techniques using Power BI and Matplotlib, presenting research findings through scatter plots, bar charts, and heatmaps, aiding stakeholder understanding and supporting data-driven decision-making.

- Collaborated with cross-functional teams to align data initiatives with broader research goals, ensuring successful implementation of data-driven solutions and improving health outcomes through advanced analytics.
- Led the AIM-AHEAD Pilot project, overseeing data management and analysis to improve health outcomes, utilizing advanced machine learning and data integration methodologies for health research.
- Contributed valuable insights to research papers, effectively communicating complex findings derived from data analysis to stakeholders and the scientific community, driving informed decisions in health policy.

DATA ENGINEER | Cognizant Technology Solutions | Chennai, IN

December 2020 - July 2022

- Led the migration of ETL job scripts from legacy tools like Informatica, IBM DataStage, and Talend to cloud-based solutions such as AWS Glue, Azure Data Factory, and Google Dataflow using Apache Beam and PySpark, optimizing data pipelines for improved performance and scalability.
- Designed and optimized scalable data pipelines using Google BigQuery, Redshift, and Azure Synapse, enhancing data ingestion and transformation processes to support real-time analytics and business intelligence.
- Spearheaded automation and optimization of ETL pipelines using Apache Airflow, reducing manual intervention and improving workflow orchestration across cloud platforms.
- Implemented CI/CD pipelines using Jenkins and Azure DevOps for automated testing, deployment, and monitoring of data engineering solutions, improving time-to-market and operational efficiency.
- Developed and maintained Databricks notebooks to orchestrate complex ETL workflows using PySpark and SQL, ensuring seamless execution and scalability of data engineering processes across cloud environments.
- Utilized dbt to streamline and automate the transformation of raw data into analytics-ready datasets within cloud-based data warehouses (e.g., BigQuery, Snowflake, Redshift), ensuring high-quality, maintainable data models by leveraging SQL-based transformations, version control, and testing frameworks for improved data reliability and faster insights.
- Built and maintained scalable data lake architectures on AWS S3, Azure Blob Storage, and GCP Cloud Storage, ensuring efficient data storage and retrieval.
- Enhanced SQL transformations and data processing workflows in Databricks by applying performance tuning techniques and leveraging advanced optimization features, resulting in quicker insights and efficient data processing.
- Automated data ingestion, transformation, and loading processes using Apache Airflow, AWS Glue, and Azure Data Factory, significantly improving operational efficiency.
- Enhanced data transformation processes by leveraging PySpark for faster data processing across large datasets, improving overall processing efficiency.
- Created and maintained dashboards and automated reporting workflows using Tableau and Power BI, delivering critical insights and supporting data-driven decision-making.
- Implemented data quality assurance practices, including data cleansing, profiling, and validation, to ensure high data accuracy and reliability for business-critical reports.
- Integrated databases like Oracle, Snowflake, Redshift, BigQuery, MongoDB, and Bigtable, optimizing performance and ensuring data integrity for various business applications.
- Containerized data engineering workflows using Docker to ensure consistent development and deployment environments, enabling scalable and portable ETL execution across cloud and on-premises systems.
- Resolved complex technical issues related to data pipelines and ETL processes, ensuring continuous data flow and minimal downtime, contributing to accelerated time-to-market.
- Conducted comprehensive knowledge transfer (KT) sessions to onboard new team members and led workshops on Git for version control and collaborative coding best practices.
- Collaborated with cross-functional teams including data scientists, analysts, and product managers to define data requirements and ensure alignment of ETL solutions with evolving business needs and analytical goals.
- Developed comprehensive documentation, including design specifications and mapping documents, to facilitate knowledge sharing and project transparency.
- Executed advanced SQL querying across databases like Oracle, Teradata, and MongoDB, enhancing data extraction and transformation processes to meet complex analytical requirements.

PROGRAMMER ANALYST | Krest Technologies | IN

May 2019 - Oct 2020

- Assisted in the design and development of scalable data pipelines using Python and SQL, ensuring seamless data flow and integration across various platforms and cloud environments.
- Developed and maintained ETL processes utilizing Apache Airflow, Informatica, and custom Python scripts to automate data extraction, transformation, and loading, enhancing operational efficiency.
- Ingested data from disparate sources using SQL and Google Analytics API, constructing data views for BI tools like Tableau to

support real-time analytics and reporting.

- Conducted data analysis to support business decision-making, leveraging SQL and Python to generate actionable insights from large datasets, driving strategic initiatives.
- Optimized query performance and improved data retrieval times by tuning SQL queries and implementing indexing strategies, resulting in faster and more efficient data access for end-users.
- Created and maintained comprehensive documentation for data pipelines, ETL processes, and data models to facilitate knowledge sharing and ensure process transparency.
- Collaborated closely with cross-functional teams, including data scientists, analysts, and business stakeholders, to gather requirements, design data models, and deliver tailored data solutions that align with business objectives.
- Developed and deployed RESTful APIs to facilitate data access and integration, enhancing the usability of data solutions for internal and external stakeholders.
- Implemented best practices in data governance and quality assurance, ensuring data accuracy and compliance with organizational standards.
- Participated in code reviews and contributed to the optimization of existing Python scripts, enhancing workflow efficiency and reliability.
- Engaged in continuous learning and knowledge sharing within the team, fostering a collaborative environment focused on innovation and improvement.

Education

MASTER OF SCIENCE IN COMPUTER and INFORMATION SCIENCE | University of North Texas | TX

GPA: 3.81

BACHELORS IN COMPUTER SCIENCE | Anurag Engineering College

GPA: 3.80

Projects

Healthcare Analytics: Predictive Modeling for Heart Disease Diagnosis

- Led a machine learning project utilizing GradientBoostingClassifier to significantly improve diagnostic accuracy for heart disease, demonstrating expertise in healthcare analytics, predictive modeling, and machine learning algorithms.
- Implemented data preprocessing techniques, feature engineering, and model evaluation to enhance model performance and drive advancements in medical diagnostics.
- Utilized Python, scikit-learn, and PySpark for data processing and analysis, ensuring scalability and efficiency in handling large healthcare datasets.

Big Data Optimization: Hadoop Performance Improvement with Metadata Jobs

- Directed the H2 Hadoop development, optimizing efficiency, reducing costs, and enhancing processing speeds by implementing metadata-related jobs.
- Demonstrated proficiency in big data processing and cloud computing, utilizing Apache Hadoop, HDFS, and YARN for distributed data storage and processing.
- Employed performance tuning techniques, such as data partitioning and indexing, to elevate resource utilization and performance standards, ensuring scalability and reliability of the Hadoop infrastructure.

Financial Data Analysis: Credit Card Management System

- Developed a credit card management system using PySpark, Hadoop, and Hive, optimizing data analysis and enhancing business efficiency in financial data processing.
- Implemented custom ETL pipelines to ingest, transform, and analyze transactional data, demonstrating expertise in big data analytics, engineering, and process optimization.
- Leveraged Apache Spark for parallel processing and distributed computing, ensuring high performance and scalability for real-time data processing.